

Chapter 1 Introduction

1.1 Motivation:

- Limited Access to Mental Health Support

Many people who need psychological support face difficulties in accessing mental health professionals due to:

- Limited availability of therapists
- Geographical distance
- High cost of sessions
- Hesitation to seek psychological help

Provide an accessible platform that connects patients with mental health support through AI-based assistance and direct communication with therapists.

- Lack of Continuous Psychological Monitoring

Patients' psychological conditions and moods can change over time, but traditional support methods may not provide continuous monitoring. This can lead to:

- Difficulty tracking mood changes
- Limited information about patient progress
- Delayed recognition of changes in psychological state

Provide mood tracking, daily challenges, progress monitoring, and weekly psychological summaries to help patients and therapists follow changes over time.

- Difficulty in Assessing the Patient's Psychological State

Traditional assessment often depends mainly on conversations and information provided by the patient. This may result in:

- Limited assessment information
- Difficulty identifying emotional changes
- Dependence on subjective responses

Support psychological assessment by analyzing text, facial expressions, and voice characteristics to provide additional indicators of the patient's emotional state.

- Lack of Privacy in Expressing Sensitive Feelings

Some patients may hesitate to express their thoughts and feelings because of concerns about:

- Privacy
- Confidentiality
- Sharing sensitive conversations with others

Provide a confidential private venting mode that allows patients to express their feelings through AI without storing or sharing the conversation with the doctor.

- Difficulty Detecting Psychological Crisis Situations

Critical statements related to self-harm or other dangerous situations may require immediate attention. Delayed detection can lead to:

- Lack of immediate intervention
- Delayed communication with the therapist
- Increased risk to the patient

Implement a crisis safety mechanism that detects high-risk statements and triggers emergency support and urgent notifications.

- Fragmented Mental Health Services

Patients may need to use different services for:

- Communicating with therapists
- Booking appointments
- Receiving prescriptions
- Tracking medications
- Monitoring psychological progress
- Accessing mental health resources

Build an integrated platform that brings different mental health support services together in one convenient environment.

- Need for Better Therapist-Patient Communication

Therapists may not always have enough information about the patient's recent condition before a session, which can lead to:

- Time spent collecting basic information
- Difficulty reviewing patient progress
- Less efficient session preparation

Provide organized patient reports, AI-generated summaries, progress analytics, and communication tools to support therapists in understanding the patient's condition and improving session efficiency.

1.2 Project Objectives:

The main objective of this project is to develop an integrated mental health support platform that improves access to psychological support and facilitates effective communication between patients and qualified therapists. The platform aims to provide AI-based psychological assistance, monitor patients' moods and progress, and provide additional indicators of their emotional state through multimodal analysis. It also seeks to enhance patient privacy, enable early identification of high-risk situations, and assist therapists through organized reports and analytical insights. Additionally, the system is designed to facilitate appointment booking, medication management, personalized mental health resources, and supportive self-care activities within a secure, user-friendly, and accessible environment.

1.3 Project Developing Methodology:

The **Agile development methodology** is well-suited for developing our mental health support platform. Here are some reasons why:

- 1. Flexibility:** The Agile approach is highly flexible, allowing the development team to adapt to changing requirements and priorities quickly. This is especially useful for our project, which includes multiple components and technologies that need continuous improvement.
- 2. Iterative Development:** The iterative development process in Agile allows each feature to be developed, tested, and refined in short cycles. This helps the team identify problems early and ensure that the platform meets the required needs.
- 3. Collaboration:** The Agile approach emphasizes communication and collaboration between team members, which is important for coordinating the development of the mobile application, web dashboard, backend, and AI-based features.
- 4. Continuous Improvement:** The Agile methodology encourages continuous testing, feedback, and improvement throughout the development process. This helps the team enhance the platform's quality, usability, and reliability.



Figure 1.1: Agile Development Cycle

1.4 Tools and Technologies:

Development Environment

- Visual Studio Code
- Visual Studio Community

AI & Machine Learning

- OpenAI API (GPT-based conversational support)
- n8n / LangChain for AI workflow integration
- PHQ-9 and GAD-7 assessment frameworks
- DeepFace / OpenCV for facial emotion analysis
- Whisper for speech-to-text
- Librosa for voice tone and emotion analysis
- EasyOCR / Google Vision API for prescription recognition

WEB(REACT.JS/ TAILWIND CSS)

- React for developing the therapist and administrator web dashboard
- Tailwind CSS for styling and responsive user interfaces
- Axios for API communication
- Three.js / Babylon.js for interactive 3D visualization
- Charting libraries for mood and progress analytics

BACKEND (PHP / LARAVEL)

- PHP and Laravel for developing the RESTful backend API
- MySQL for database management
- Laravel Sanctum / JWT for authentication and API security
- n8n for automated workflows and scheduled notifications
- Google Maps Places API for pharmacy location services
- Paymob / Fawry for payment integration

MOBILE (FLUTTER)

- Flutter for developing the patient mobile application
- Dart programming language
- Flutter audio player libraries for calming sounds
- flutter_local_notifications for medication reminders
- Firebase Cloud Messaging (FCM) for notifications
- Google Maps and Geolocation services for pharmacy locations

1.5 Definitions, Acronyms, and Abbreviations:

- **API** : Application Programming Interface
- **LLM** : Large Language Model
- **OCR** : Optical Character Recognition
- **PHQ-9** : Patient Health Questionnaire-9
- **GAD-7** : Generalized Anxiety Disorder-7
- **JWT** : JSON Web Token
- **FCM** : Firebase Cloud Messaging